IT314 – Lab Assignment 8

Test Results: Home Appliance Repair Booking System

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Test execution overview

The program was compiled with `javac -Xlint:all` and executed using the automated test driver. All seven required scenarios were run against the same transition rules documented in the requirements report and UML diagram. The full console output below is the actual execution evidence.

Execution environment

Item	Detail
Language/runtime	Java 21 or newer
Compiler	`javac -Xlint:all`
Test command	`java Main --tests`
Dependencies	None; standard Java library only
Test scope	TC01 through TC07

Results summary

Test	Expected	Actual final state / observation	Result
TC01	Successful payment, center acceptance, and completion	`SERVICE_COMPLETED`; all valid transitions executed	PASS
TC02	Payment failure does not mark paid; retry is possible	First attempt failed, second succeeded; `CENTER_REVIEW`	PASS
TC03	Paid rejection prevents technician assignment and refunds	`BOOKING_REJECTED`; full refund; assignment rejected	PASS
TC04	Invalid event rejected without state corruption	`PAYMENT_PENDING` unchanged after `COMPLETE_SERVICE`	PASS
TC05	Pre-assignment cancellation follows refund policy	`CANCELLED`; full refund	PASS
TC06	Cancellation after start/completion rejected	`SERVICE_IN_PROGRESS` and `SERVICE_COMPLETED` unchanged	PASS
TC07	Complete technician/service sequence	`SERVICE_COMPLETED` through all required intermediate states	PASS

Boundary condition

The cancellation boundary is demonstrated in TC06: cancellation is rejected in `SERVICE_IN_PROGRESS` and `SERVICE_COMPLETED`, while TC05 demonstrates that cancellation in `CENTER_REVIEW` is accepted with a full refund. This is correct because the policy allows cancellation only before technician assignment. The payment retry boundary is also implemented: at most three attempts are allowed; `RETRY_PAYMENT` is rejected after the third attempt.

Actual Java execution evidence

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TC01 - Successful Booking
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Expected: Payment succeeds, center accepts, and service reaches SERVICE_C
  BOOK_SERVICE | BOOKING_CREATED → PAYMENT_PENDING | Booking details acc
  PAY | PAYMENT_PENDING → CENTER_REVIEW | Payment successful (attempt 1/
  ACCEPT_BOOKING | CENTER_REVIEW → BOOKING_CONFIRMED | Service center ac
  ASSIGN_TECHNICIAN | BOOKING_CONFIRMED → TECHNICIAN_ASSIGNED | Technici
  DISPATCH_TECHNICIAN | TECHNICIAN_ASSIGNED → TECHNICIAN_EN_ROUTE | Techn
  START_SERVICE | TECHNICIAN_EN_ROUTE → SERVICE_IN_PROGRESS | Repair ser
  COMPLETE_SERVICE | SERVICE_IN_PROGRESS → SERVICE_COMPLETED | Repair co
Actual: All valid transitions completed; paid=true | Final State: SERVICE_
Result: PASS

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TC02 - Payment Failure and Retry
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Expected: First payment fails, retry succeeds, and booking becomes CENTER_
  BOOK_SERVICE | BOOKING_CREATED → PAYMENT_PENDING | Booking details acc
  PAY | PAYMENT_PENDING → PAYMENT_FAILED | Payment failed (attempt 1/3);
  RETRY_PAYMENT | PAYMENT_FAILED → PAYMENT_PENDING | Retry permitted; at
  PAY | PAYMENT_PENDING → CENTER_REVIEW | Payment successful (attempt 2/
Actual: Failure preserved unpaid state; retry succeeded | Final State: CEN
Result: PASS

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TC03 - Service-Center Rejection
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Expected: Paid booking is rejected and fully refunded; technician cannot b
  BOOK_SERVICE | BOOKING_CREATED → PAYMENT_PENDING | Booking details acc
  PAY | PAYMENT_PENDING → CENTER_REVIEW | Payment successful (attempt 1/
  REJECT_BOOKING | CENTER_REVIEW → BOOKING_REJECTED | Full simulated refu
  INVALID TRANSITION: ASSIGN_TECHNICIAN is not allowed from BOOKING_REJECT
Actual: Rejected booking stayed terminal and refund=true | Final State: BO
Result: PASS

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TC04 - Invalid Transition
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Expected: COMPLETE_SERVICE from PAYMENT_PENDING is rejected and state rema
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  BOOK_SERVICE | BOOKING_CREATED → PAYMENT_PENDING | Booking details acc
  INVALID TRANSITION: COMPLETE_SERVICE is not allowed from PAYMENT_PENDIN
Actual: State unchanged after invalid event | Final State: PAYMENT_PENDIN
Result: PASS

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TC05 - Cancellation Before Technician Assignment
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Expected: Paid booking cancels during CENTER_REVIEW with a full simulated
  BOOK_SERVICE | BOOKING_CREATED → PAYMENT_PENDING | Booking details acc
  PAY | PAYMENT_PENDING → CENTER_REVIEW | Payment successful (attempt 1/3
  CANCEL | CENTER_REVIEW → CANCELLED | Cancellation accepted before techn
Actual: Cancellation accepted before assignment; full refund=true | Final
Result: PASS

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TC06 - Cancellation After Service Start / Completion
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Expected: Cancellation is rejected after service starts and after completi
  BOOK_SERVICE | BOOKING_CREATED → PAYMENT_PENDING | Booking details acc
  PAY | PAYMENT_PENDING → CENTER_REVIEW | Payment successful (attempt 1/3
  ACCEPT_BOOKING | CENTER_REVIEW → BOOKING_CONFIRMED | Service center acc
  ASSIGN_TECHNICIAN | BOOKING_CONFIRMED → TECHNICIAN_ASSIGNED | Technicia
  DISPATCH_TECHNICIAN | TECHNICIAN_ASSIGNED → TECHNICIAN_EN_ROUTE | Techn
  START_SERVICE | TECHNICIAN_EN_ROUTE → SERVICE_IN_PROGRESS | Repair serv
  INVALID TRANSITION: CANCEL is not allowed from SERVICE_IN_PROGRESS. Stat
  BOOK_SERVICE | BOOKING_CREATED → PAYMENT_PENDING | Booking details acc
  PAY | PAYMENT_PENDING → CENTER_REVIEW | Payment successful (attempt 1/3
  ACCEPT_BOOKING | CENTER_REVIEW → BOOKING_CONFIRMED | Service center acc
  ASSIGN_TECHNICIAN | BOOKING_CONFIRMED → TECHNICIAN_ASSIGNED | Technicia
  DISPATCH_TECHNICIAN | TECHNICIAN_ASSIGNED → TECHNICIAN_EN_ROUTE | Techn
  START_SERVICE | TECHNICIAN_EN_ROUTE → SERVICE_IN_PROGRESS | Repair serv
  COMPLETE_SERVICE | SERVICE_IN_PROGRESS → SERVICE_COMPLETED | Repair com
  INVALID TRANSITION: CANCEL is not allowed from SERVICE_COMPLETED. State
Actual: Both post-boundary cancellations rejected without state corruption
Result: PASS

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TC07 - Complete Service Flow
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Expected: Valid sequence reaches SERVICE_COMPLETED via all technician/serv
  BOOK_SERVICE | BOOKING_CREATED → PAYMENT_PENDING | Booking details acc
  PAY | PAYMENT_PENDING → CENTER_REVIEW | Payment successful (attempt 1/3
  ACCEPT_BOOKING | CENTER_REVIEW → BOOKING_CONFIRMED | Service center acc
```

```
  ASSIGN_TECHNICIAN | BOOKING_CONFIRMED → TECHNICIAN_ASSIGNED | Technicia
  DISPATCH_TECHNICIAN | TECHNICIAN_ASSIGNED → TECHNICIAN_EN_ROUTE | Techn
  START_SERVICE | TECHNICIAN_EN_ROUTE → SERVICE_IN_PROGRESS | Repair serv
  COMPLETE_SERVICE | SERVICE_IN_PROGRESS → SERVICE_COMPLETED | Repair com
Actual: TECHNICIAN_ASSIGNED → TECHNICIAN_EN_ROUTE → SERVICE_IN_PROGRESS
Result: PASS

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OVERALL RESULT: PASS
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Overall result: PASS.